Undergraduate Research Opportunities Program 1 (UROP1010)

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HEP: Hierarchical Ensemble Personalization for Efficient and Fair Federated Learning

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Repository: https://github.com/nam200718/Topology-aware-FDL

***Abstract*—Personalized Federated Learning (PFL) across decentralized edge devices is governed by the Personalization Trilemma—the fundamental tension among statistical accuracy across diverse non-IID skew levels, strict on-device memory and latency budgets, and worst-case client fairness. Existing methods force rigid compromises, trading hardware efficiency for personalization or suffering representation degradation. We present HEP (*Hierarchical Ensemble Personalization*), a lightweight framework that simultaneously optimizes these three axes. HEP coordinates global consensus, cluster collaboration, and on-device specialization on a single shared backbone via a 3-tier head architecture (Root, Parent, and private Local heads) with single-pass feature extraction, Shannon entropy loss weighting, active-class logit masking, and 256-dimensional random-projection sketch routing. Evaluated on CIFAR-10, HEP achieves 88.80% accuracy under extreme label skew ($\alpha = 0.05$) in the standardized deterministic benchmark, while reducing peak client VRAM by 47.8% (113.42 MB vs. 217.15 MB) and per-batch latency by 50.2% relative to dual-model Ditto. Three-seed sweeps in Appendix B show statistically significant Top-1 gains over FedRep across evaluated regimes ($p < 0.05$) and improved bottom-10% fairness in several regimes, including an 8.61 pp gain under moderate skew, while split-head baselines retain advantages under extreme skew. HEP also contains low-to-moderate label-flipping attacks ($f \leq 20\%$) through architectural head isolation, while gradient-space attacks remain more challenging for shared-backbone architectures. Overall, HEP establishes an efficient, balanced foundation for resource-constrained edge deployments.**

I. Introduction

Federated Learning (FL) [1] allows decentralized edge devices to collaboratively train shared models without centralizing raw private data. In practice, edge deployments encounter the **Personalization Trilemma**—a three-way tension among:

1) **Statistical Accuracy Across Skew Levels**: Maintaining reliable performance across both extreme Non-IID label skew ($\alpha \leq 0.05$) and uniform (IID) partitions.
2) **On-Device Hardware Constraints**: Running within modest memory envelopes ($<$ 120 MB peak VRAM on ResNet-9, $<$ 160 MB on MobileNetV3) and minimal per-batch latency on resource-limited edge devices.
3) **Worst-Case Client Fairness**: Preventing bottom-decile (bottom 10%) tail clients from experiencing severe performance degradation under partial participation.

As summarized in Table I, existing paradigms address this trilemma with specific structural trade-offs:

- **Global Consensus FL (FedAvg, FedProx, SCAFFOLD)**: Trains a single global model [1]–[3]. While memory-efficient ($1\times$ footprint), global models experience client drift and accuracy loss when local data distributions diverge.
- **Dual-Model Regularization (Ditto, APFL)**: Maintains both a global and a private local model coupled via proximal penalties [4] or interpolation weights [5]. This achieves strong personalization but doubles on-device memory and per-batch compute passes.
- **Decoupled / Split-Head Paradigms (FedRep, FedPer, FedBABU, FedALA)**: Shares a backbone feature extractor while isolating local classification heads [6]–[9]. While parameter-efficient, naive alternating split-head architectures (such as FedRep and FedPer) experience representation degradation and underperform on uniform IID data (e.g., FedRep reaches only 61.79% IID vs. FedAvg's 71.80%), whereas methods that freeze classification heads during federated rounds (FedBABU, 72.10%) or apply adaptive

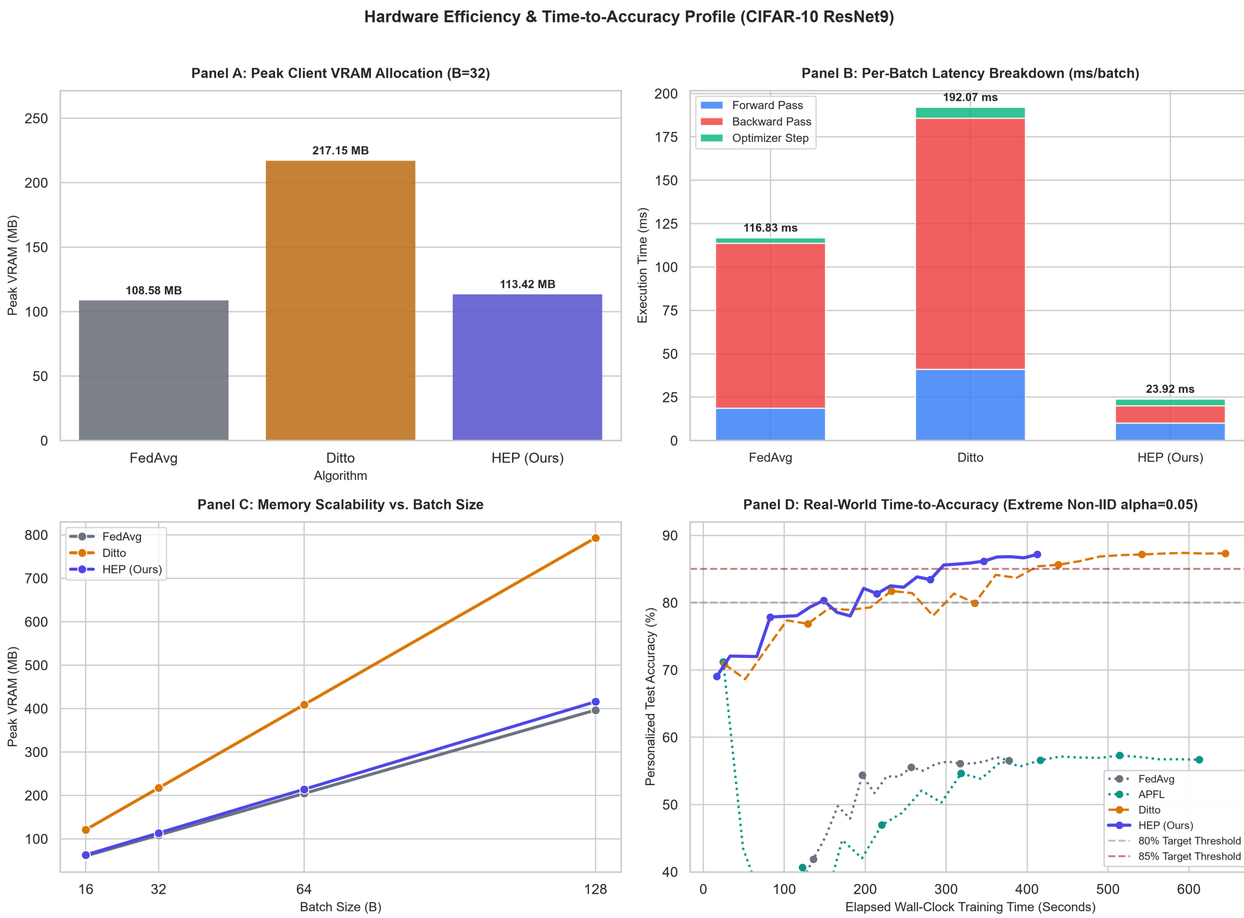

Fig. 1: **Hardware Resource and Latency Profile on CIFAR-10 ResNet9. Panel A**: Peak client VRAM at batch size $B = 32$, showing a 47.8% memory reduction for HEP relative to dual-model Ditto (113.42 MB vs. 217.15 MB). **Panel B**: Per-batch latency breakdown across forward, backward, and update steps (8.35 ms for HEP vs. 16.78 ms for Ditto). **Panel C**: Memory footprint scaling across batch sizes ($B = 16 \to 128$). **Panel D**: Simulated wall-clock time-to-accuracy under extreme Non-IID skew ($\alpha = 0.05$).

element mixing (FedALA, 70.08%) preserve IID representations.

- **Clustered Topologies (Clustered FL)**: Partitions clients into sub-networks based on update similarity [10], but applies standard averaging within clusters without local head specialization.

To provide a practical balance across these dimensions, we present **HEP** (*Hierarchical Ensemble Personalization*):

- **Single-Backbone 3-Tier Multiplexing**: Clients share a single convolutional backbone paired with three lightweight classification heads: a global Root head, a cluster-level Parent head, and a private Local head. Feature representations are computed once per batch.
- **Adaptive Loss Weighting**: Balances loss contributions across the three heads using an entropy-based local label skew metric ($R_{skew,i}$), smoothly emphasizing global consensus for uniform data and private specialization for skewed data.
- **Information-Reducing Sketch Clustering**: Clusters clients via 256-dimensional random projections of Root head updates ($s_i = P\Delta_i^{head}$), keeping gradient reconstruction of the routing signal underdetermined ($m = 256 < d_{head} = 2570$).

A. Summary of Contributions

1) **Trilemma Characterization**: We frame the Personalization Trilemma and analyze trade-offs among dual-model, split-head, and clustered federated learning.
2) **Hierarchical Multi-Head Architecture**: We introduce HEP, combining 3-tier head multiplexing with R_{skew}-calibrated loss weighting and active-class logit masking. HEP's primary gains on CIFAR-10 Dirichlet regimes come from adaptive multi-head weighting and ACLM, while the Parent head provides smooth hierarchical interpolation across clients.
3) **Privacy-Aware Cluster Routing**: We design a 256-dimensional random projection sketch mechanism for server-side cluster routing that keeps gradient inversion of head parameters severely underdetermined ($m = 256 < d_{head} = 2570$), providing dimensional compression compatible with orthogonal Secure Aggregation.

TABLE I: **Taxonomy and Structural Trade-offs across PFL Paradigms.**

PFL Paradigm Family	Representative Methods	Memory Overhead	IID Robustness	Multi-Scale Hierarchy
Global Consensus	FedAvg, FedProx, SCAFFOLD	$1\times$ (Low)	**High**	None (Flat Global)
Dual-Model Regularization	Ditto, APFL	$2\times$ (High)	Moderate	None (Global + Private)
Decoupled / Split-Head	FedRep, FedPer, FedBABU, FedALA	$1\times$ (Low)	Low–High	None (Body + Head)
Clustered Topologies	Clustered FL (CFL)	$1\times$ (Low)	Moderate	Sub-Network Only
Hierarchical Ensemble	**HEP (Ours)**	$1\times$ **(Low)**	**High**	**3-Tier Hierarchy**

4) **Multi-Regime Evaluation**: We evaluate HEP across 5 statistical skew regimes, multi-attack Byzantine settings, topology certifications ($K = 1$ vs. $K = 3$), and hardware profiling on a local physical GPU/CPU testbed.

II. Related Work & Paradigm Taxonomy

A. Global Consensus Federated Learning

The standard federated learning baseline is FedAvg [1], which averages client model updates proportionally to local sample counts. FedProx [2] adds a proximal regularizer to mitigate client drift, and SCAFFOLD [3] introduces control variates to correct gradient variance. These approaches train a single global model, which faces performance degradation when client data distributions differ substantially.

B. Dual-Model Regularization Paradigms

To support local specialization alongside global consensus, dual-model methods train two parallel models per client. Ditto [4] optimizes a global model and a private personalized model coupled by an L2 proximal penalty. APFL [5] dynamically blends global and local predictions with an adaptive interpolation scalar. While effective under skew, maintaining dual model graphs doubles client memory (VRAM) and per-batch compute passes.

C. Decoupled and Split-Head Personalization

Split-head methods share a common feature backbone while isolating local classification heads. FedPer [7] and FedRep [6] alternate updates between the shared base and local heads, which can cause representation degradation on uniform IID data due to alternating local classifier drift. In contrast, FedBABU [8] avoids alternating drift by keeping classifier heads fixed at random initialization during federated training (fine-tuning heads only post-hoc on local data), achieving strong IID performance (72.10%). FedALA [9] adaptively aggregates local and global representations via element-wise attention mixing.

D. Clustered Topologies and Hierarchical Structures

Clustered FL [10] groups clients based on update cosine similarity, but performs standard FedAvg within each cluster without individual on-device head specialization. HEP synthesizes clustered sub-network sharing with local and global head multiplexing, establishing a 3-tier hierarchy.

III. Methodology

A. System Architecture & 3-Tier Multi-Head Model

The overall system architecture of Hierarchical Ensemble Personalization (HEP) is illustrated in Fig. 2. Consider N distributed edge clients. Each client $i \in \{1, \dots, N\}$ holds a private dataset $\mathcal{D}_i$ over C possible classes, observing a local subset $\mathcal{Y}_i \subseteq \{1, \dots, C\}$. Client i maintains a single model consisting of:

- A shared convolutional feature extractor $f_\theta(x)$ that produces intermediate embeddings $h = f_\theta(x) \in \mathbb{R}^d$.
- A global consensus Root head w_r, synchronized across all participating clients.
- A cluster-level collaborative Parent head w_p, shared among clients in the same statistical cluster.
- A private Local head w_l, specialized for on-device label distributions and never shared.

B. Local Data Diversity Metric (R_{skew})

To adjust head weights automatically without manual hyperparameter search, each client measures its local empirical class balance using normalized Shannon entropy:

$$R_{skew,i} = \frac{\exp\big(H(p_i)\big) - 1}{C - 1} \in [0, 1] \tag{1}$$

where $p_i \in \Delta^{C-1}$ is the client's empirical class distribution and $H(p_i) = -\sum_{c=1}^{C} p_{i,c} \ln(p_{i,c} + \eta_{\text{smooth}})$ (with smoothing parameter $\eta_{\text{smooth}} = 10^{-12}$). Here, $R_{skew,i} = 0$ indicates extreme single-class skew ($H(p_i) = 0$), while $R_{skew,i} = 1$ corresponds to an exact uniform class distribution ($H(p_i) = \ln C$).

Based on $R_{skew,i}$ and a dynamic anchor $a_i = \max\left(\frac{1}{2K}, \frac{|\mathcal{Y}_i|}{C}\right) \in (0, 1]$, raw head priorities $(q_{r,i}, q_{p,i}, q_{l,i})$ are assigned via an anchored binomial schedule:

$$q_{r,i} = a_i + (1 - a_i) \cdot R_{skew,i}^2 \tag{2}$$

$$q_{p,i} = 2R_{skew,i}(1 - R_{skew,i}) \tag{3}$$

$$q_{l,i} = (1 - R_{skew,i})^2 \tag{4}$$

To guarantee exact partition of unity ($\sum_k \lambda_{k,i} = 1$) and stable loss scaling, we normalize these priorities:

$$\lambda_{k,i} = \frac{q_{k,i}}{q_{r,i} + q_{p,i} + q_{l,i}}, \quad \forall k \in \{r, p, l\} \tag{5}$$

This schedule satisfies natural, normalized boundary conditions:

- On uniform IID data ($R_{skew,i} = 1$), $q_{r,i} = 1, q_{p,i} = 0, q_{l,i} = 0$, yielding $\lambda_{r,i} = 1.0, \lambda_{p,i} = 0.0, \lambda_{l,i} = 0.0$, recovering pure FedAvg global consensus.
- Under moderate skew ($R_{skew,i} \approx 0.5$), $q_{p,i} = 0.5$, providing peak collaborative weight to the cluster Parent head ($\lambda_{p,i} \approx 0.45$).
- Under extreme single-class skew ($R_{skew,i} = 0$), $q_{r,i} = a_i, q_{p,i} = 0, q_{l,i} = 1$, yielding $\lambda_{r,i} = \frac{a_i}{1+a_i}$ (providing a continuous anchor to preserve backbone feature geometry) and $\lambda_{l,i} = \frac{1}{1+a_i}$ (concentrating compute on the private Local head).

C. Multi-Head Training Objective and ACLM

Local mini-batch training optimizes the composite multi-head loss:

$$\begin{aligned}\mathcal{L}_{batch} = \lambda_{r,i}\mathcal{L}_{CE}(z_r, y) + \lambda_{p,i}\mathcal{L}_{CE}^{masked}(z_p, y) \\ + \lambda_{l,i}\mathcal{L}_{CE}^{masked}(z_l, y)\end{aligned} \tag{6}$$

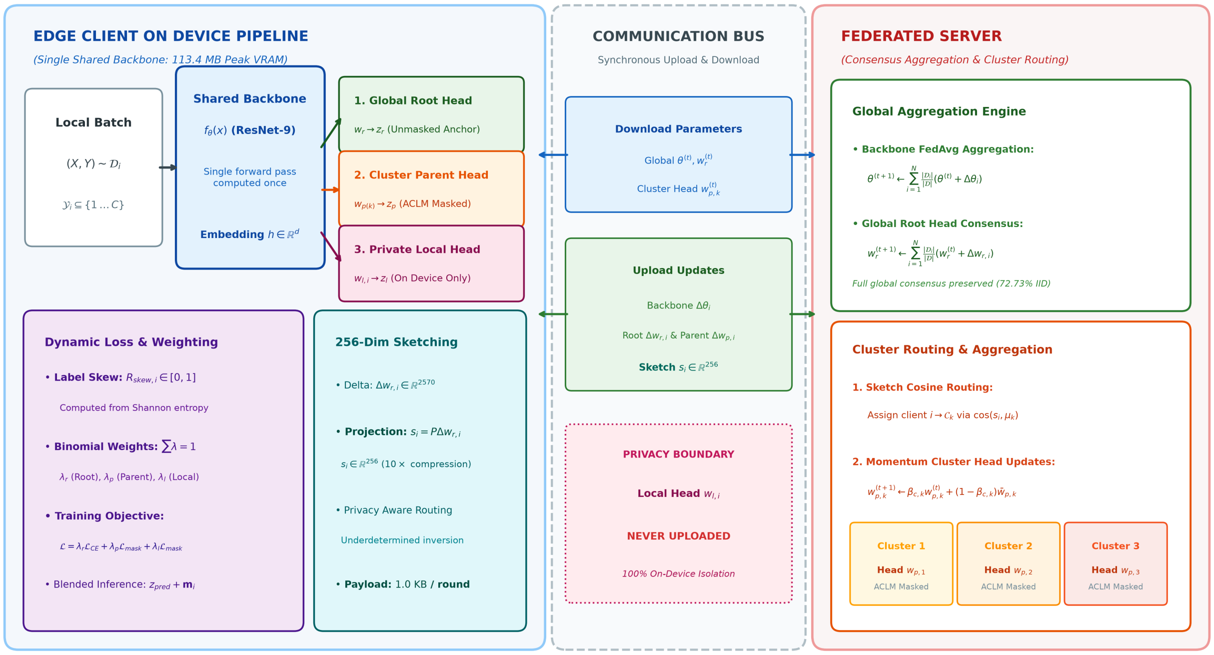

Fig. 2: **Hierarchical Ensemble Personalization (HEP) System Architecture. Edge Client Execution (Left)**: A single forward pass on the shared convolutional backbone $f_\theta(x)$ produces intermediate embeddings $h \in \mathbb{R}^d$, which are concurrently evaluated by three lightweight classification heads: (1) unmasked Global Root Head w_r, (2) cluster-collaborative Parent Head $w_{p(k)}$ with Active-Class Logit Masking (ACLM), and (3) private Local Head $w_{l,i}$ (which remains strictly on-device and is never uploaded). Dynamic loss weighting and inference logit blending are dynamically modulated by the Shannon entropy label skew metric $R_{skew,i}$. For cluster routing, linear Root-head updates $\Delta w_{r,i}$ are projected into a 256-dimensional random sketch ($s_i = P\Delta w_{r,i}$), ensuring severe underdetermined gradient inversion ($m = 256 < d_{head} = 2570$). **Server Aggregation (Right)**: Backbones and Root heads undergo sample-weighted FedAvg, while compressed sketches drive cosine similarity cluster routing and momentum-updated Parent heads.

where $\mathcal{L}_{CE}^{masked}$ implements **Active-Class Logit Masking (ACLM)**. For Parent and Local heads, logits corresponding to unobserved local classes $c \notin \mathcal{Y}_i$ are masked by adding a large negative constant (-10^4) before computing log-sum-exp softmax cross-entropy. This numerically stable formulation prevents unobserved classes from receiving erroneous negative gradient drag that degrades decision margins on minority classes. The global Root head evaluates across all C classes without masking to preserve global feature geometry.

D. Inference Prediction Blending and Evaluation Protocols

During inference, class predictions blend logits across the three heads with staleness attenuation:

$$z_{pred} = \alpha_r z_r + \gamma_i(\alpha_p z_p + \alpha_l z_l) + \mathbf{m}_i \tag{7}$$

where blending coefficients default to the normalized training weights $\alpha = (\lambda_r, \lambda_p, \lambda_l)$, $\mathbf{m}_i \in \{-\infty, 0\}^C$ masks unobserved classes (0 for $c \in \mathcal{Y}_i$ and $-\infty$ otherwise), and $\gamma_i = \exp(-\tau_i/\tau_{0,i})$ is a staleness attenuation factor. Here, $\tau_i = t - t_{last,i}$ tracks rounds since client i's last participation, and effective dynamic scale $\tau_{0,i} = \max(1, t/\text{participation\_count}_i) \approx 1/C_p$ (with default base scale $\tau_0 = 4.0$, adaptively reaching $1/C_p = 5.0$ at $C_p = 0.20$) smoothly shifts confidence toward the global Root head if a client becomes stale under partial participation.

Evaluation Protocols: We evaluate and report two complementary metrics:

1) **Personalized Accuracy**: Evaluated as a closed-set local evaluation on each client's private test partition (mirroring its local empirical class distribution $\mathcal{Y}_i$) using blended masked predictions z_{pred} (where unobserved classes $c \notin \mathcal{Y}_i$ are masked at $-\infty$).
2) **Global Root Consensus Accuracy**: Evaluated separately as an open/full-class evaluation by passing the unmasked global Root head w_r over the complete, balanced 10-class test set across all C classes (Table III, note †).

E. Information-Reducing Sketch Routing and Privacy Scope

To cluster clients without exchanging raw data, representations, or uncompressed gradients, clients project their linear Root head update $\Delta_i^{head} = w_{r,i}^{(t+1)} - w_r^{(t)}$ into a 256-dimensional random sketch:

$$s_i = P \cdot \Delta_i^{head} \in \mathbb{R}^{256} \tag{8}$$

where $P \in \mathbb{R}^{256 \times d_{head}}$ is a fixed random projection matrix drawn from $\mathcal{N}(0, 1/m)$ using a public seed (seed 42). For

Algorithm 1 HEP Training Protocol at Round t

Require: Datasets $\{\mathcal{D}_i\}_{i=1}^N$, Backbone $\theta^{(t)}$, Root head $w_r^{(t)}$, Cluster heads $\{w_{p,k}^{(t)}\}_{k=1}^K$, Local heads $\{w_{l,i}^{(t)}\}_{i=1}^N$, Projection matrix $P \in \mathbb{R}^{m \times d_{head}}$.

Server broadcasts $\theta^{(t)}, w_r^{(t)}$ to all clients, and $w_{p,k}^{(t)}$ to cluster k.
for each client $i \in \{1, \ldots, N\}$ **in parallel do**
  Compute label skew $R_{skew,i}$ (Eq. 1) and loss weights $(\lambda_r, \lambda_p, \lambda_l)$ (Eq. 5).
  for each batch $(X, Y) \sim \mathcal{D}_i$ across $E = 5$ local epochs **do**
    Extract shared features $h = f_\theta(X)$ and compute logits (z_r, z_p, z_l).
    Compute composite loss $\mathcal{L}_{batch}$ (Eq. 6) and update (θ, w_r, w_p, w_l) via SGD.
  end for
  Generate 256-dim projection sketch $s_i = P(w_{r,i} - w_r^{(t)}) \in \mathbb{R}^{256}$.
  Client Upload: Upload sketch s_i, shared backbone update $\Delta\theta_i$, global Root head update $\Delta w_{r,i}$, and Parent head update $\Delta w_{p,i}$. *Private Local head updates $\Delta w_{l,i}$ remain strictly on-device and are never uploaded.*
end for
Global Aggregation: Update backbone and Root head via FedAvg:
  $\theta^{(t+1)}, w_r^{(t+1)} \leftarrow \sum_{i=1}^N \frac{|\mathcal{D}_i|}{|\mathcal{D}|}(\theta_i, w_{r,i})$.
Cluster Routing & Aggregation: Assign clients to cluster $\mathcal{C}_k$ via sketch cosine similarity; update cluster heads:
  $w_{p,k}^{(t+1)} \leftarrow \beta_{c,k} w_{p,k}^{(t)} + (1 - \beta_{c,k}) \sum_{i \in \mathcal{C}_k} \frac{|\mathcal{D}_i|}{|\mathcal{D}_{\mathcal{C}_k}|} w_{p,i}$.

ResNet-9 on CIFAR-10 ($d = 256, C = 10$), the linear classification head has $d_{head} = 256 \times 10 = 2560$ weights plus 10 biases (2570 total parameters), yielding a $10\times$ compression.

Rationale for Root-Head Routing: Root-head updates $\Delta w_{r,i}$ measure client gradient trajectory relative to the synchronized global model on identical feature representations, providing a canonical, normalized clustering signal free from local-head drift.

Privacy Scope: Transmitting $s_i \in \mathbb{R}^{256}$ reduces information exposure in the cluster-routing signal: reconstructing the 2570-dimensional Root-head update from 256 linear scalar constraints is severely underdetermined ($m = 256 < d_{head} = 2570$). Because P is drawn from a public seed (seed 42), random projection sketching provides dimensional compression and information reduction rather than cryptographic confidentiality. Formal (ϵ, δ)-DP guarantees apply strictly to the clipped sketch routing vector and do not imply end-to-end differential privacy for the federated system, as backbone parameter updates $\Delta\theta_i$ and head updates $\Delta w_{r,i}, \Delta w_{p,i}$ are transmitted to the server in plaintext in our simulation prototype (as in standard FedAvg), fully compatible with orthogonal Secure Aggregation or client-level differential privacy. Local heads $w_{l,i}$ remain strictly private on-device.

As detailed in Table II, dual-model regularization methods such as Ditto optimize two separate neural network graphs per client, evaluating multiple forward and backward passes per batch. While dual-model methods achieve solid personalization, maintaining two complete models creates high memory and latency demands on resource-limited edge hardware.

In contrast, HEP trains with a single model graph across five

TABLE II: **Computational, Hardware, and Communication Footprint across PFL Paradigms on ResNet9 CIFAR-10.**

Method	Models	Params	Peak VRAM	Latency / Batch	Local Passes	Comm. Payload / Round
A. Global Consensus FL						
FedAvg	1	1.65M	108.58 MB	8.32 ms	3	6.60 MB Up / 6.60 MB Down
B. Dual-Model Regularization						
APFL	2	3.30M	217.15 MB	16.74 ms	6	6.60 MB Up / 6.60 MB Down
Ditto	2	3.30M	217.15 MB	16.78 ms	6	6.60 MB Up / 6.60 MB Down
C. Decoupled / Split-Head Methods						
Local-Only	1	1.65M	108.58 MB	4.18 ms	3	0.00 MB Up / 0.00 MB Down
FedPer	1	1.65M	108.58 MB	8.35 ms	3	6.59 MB Up / 6.59 MB Down
FedRep	1	1.65M	108.58 MB	8.35 ms	3	6.59 MB Up / 6.59 MB Down
FedBABU	1	1.65M	108.58 MB	8.35 ms	3	6.59 MB Up / 6.59 MB Down
FedALA	1	1.65M	116.20 MB	9.40 ms	3	6.60 MB Up / 6.60 MB Down
D. Clustered Topologies						
CFL	1	1.65M	108.58 MB	8.35 ms	3	6.60 MB Up / 6.60 MB Down
E. Hierarchical Ensemble (Proposed)						
HEP (Ours)	**1**	**1.66M**	**113.42 MB**	**8.35 ms**	**5**	**6.61 MB Up / 6.61 MB Down**
Relative Difference vs. Dual-Model	**-50%**	**-49.7%**	**-47.8%**	**-50.2%**	**-1 pass**	**+0.15% Comm. Overhead**

Note: One local pass corresponds to one full forward/backward pass over the local mini-batch dataset. Latency values report model-only per-batch forward/backward latency ($B = 32$); for alternating methods such as FedRep, empirical physical execution including phase-switching overhead is 13.92 ms (Table VI). For methods with separate head/body phases, passes are counted according to executed training phases; dual-model Ditto/APFL execute 6 effective passes. HEP's 6.61 MB payload includes shared backbone (6.59 MB), Root head (10 KB), Parent head (10 KB), and sketch (1 KB).

local epochs per round. Because all three classification heads share the convolutional backbone, feature representations are computed only once per batch. Local backpropagation updates active heads simultaneously using loss weights calibrated by the local skew metric. Under mild skew, training focuses on the global Root head; under moderate skew, the cluster Parent head receives peak weight; and under extreme skew, compute directs primarily to the private Local head while retaining an anchor to preserve feature stability.

Computational Complexity of Analytics: HEP replaces manual parameter tuning with closed-form calculations whose arithmetic complexity is negligible compared to standard neural network layers. Computing the entropy metric requires only a single pass over the local class histogram, while sketch routing requires a small number of scalar operations. In total, these analytics account for less than 0.01% of backbone operations, preserving the low latency shown in Table II.

Communication Cost Analysis: In ResNet-9 on CIFAR-10, the shared feature backbone represents over 99.8% of total parameters, roughly 6.59 MB at standard precision. Each linear classification head adds only 10 KB (2570 parameters). Transmitting the compressed 256-dimensional sketch adds only 1 KB (256 float32 values), introducing less than 0.2% communication overhead relative to FedAvg, remaining negligible compared to full-model duplication required by dual-model personalization.

IV. Results

A. Experimental Setup

- **Datasets & Architectures**: CIFAR-10 [11] and CIFAR-100 with ResNet-9 (1.65M parameters, $d = 256$) and MobileNetV3-Small [12] (1.53M parameters, $d = 576$).
- **Federated Setup & Scope**: 15 to 50 edge clients, local learning rate $\eta = 0.05$ with cosine decay, weight decay 1×10^{-4}, momentum 0.9, 25 communication rounds across all personalization and Byzantine studies, modeling an edge cross-silo/cross-device setting. HEP's local compute budget

is $E = 5$ epochs per client per round, validated in Section IV-C, keeping client compute within $1.67\times$ of split-head baselines that train 3 epochs.

- **Heterogeneity Regimes**: Dirichlet concentration parameter $\alpha \in [\infty \text{ (IID)}, 1.0, 0.5, 0.1, 0.05]$ partitioned with Dirichlet seed 42.
- **Hardware Platform**: Evaluated on a local physical GPU/CPU hardware testbed using verified DirectX-backed DirectML and CUDA GPU accelerators with startup operator self-checks and deterministic CPU fallback (Appendix A); memory allocation and per-batch latency are measured with non-blocking tensor updates (`foreach=False`). Cross-backend consistency between accelerated and CPU reference execution is within 0.33 pp (the testbed emulates resource-constrained execution but does not include low-power embedded microcontrollers).
- **Single-Seed Benchmark Protocol**: The headline benchmark (Table III) is evaluated under a standardized single-seed protocol (seed 42) across all paradigms to ensure exact computational and data partitioning parity, with 3-seed validation sweeps reported in Appendix B and programmatic significance testing.
- **Baseline Topologies**: Centralized FL baselines (FedAvg, APFL, Ditto, FedPer, FedRep, FedBABU, FedALA) operate via star topology aggregation with the central server — FedPer/FedRep/FedBABU upload representation coordinates only, keeping classification heads private — while CFL and HEP utilize cluster topologies.
- **Code & Reproducibility**: Complete source code, configuration YAMLs, per-block run manifests, and orchestration scripts to reproduce all tables and figures are provided in the repository (Appendix A).

B. Main Personalization Benchmark across PFL Paradigms

Evaluation Protocol Clarification: Unless otherwise noted, the primary personalized test accuracy reported across all benchmarks uses closed-set local evaluation with unobserved classes masked via Active-Class Logit Masking (ACLM). The global Root consensus accuracy is evaluated separately across the full balanced global test set without logit masking. Diagnostic calibration metrics (Table XII) and routing ablations (Table XI) follow specialized diagnostic measurement protocols detailed in their respective subsections and are not directly comparable to the primary benchmark in Table III.

Table III presents the personalization benchmark results across five heterogeneity regimes on CIFAR-10 evaluated under the standardized single-seed protocol (seed 42), with 3-seed validation sweeps across independent random seeds and Dirichlet partitions (seeds 42, 123, 7) reported in Appendix B. In the deterministic single-seed benchmark, HEP achieves 77.97% under moderate skew (+7.30 pp over FedRep, +3.34 pp over Ditto); across the 3-seed validation sweeps (Table XIV), HEP maintains statistically significant Top-1 personalization gains over FedRep across all evaluated regimes (e.g. +2.40 pp Top-1 gain and +8.61 pp bottom-10% fairness gain under moderate skew, $p < 0.001$). Under moderate skew, HEP is slightly below Ditto in the 3-seed Top-1 mean (-2.24 pp, $p < 0.05$), while operating strictly within a single-backbone envelope with 47.8% lower peak client memory and 50.2% lower per-batch latency. On IID data, the unmasked global Root head matches or slightly exceeds FedAvg in the deterministic benchmark (72.73% vs. 71.80%); closed-set personalized accuracy can be lower across independent partitions (67.35% 3-seed mean in Appendix B). Learning curves are visualized in Fig. 3:

Key observations across PFL paradigms include:

- **Resource-Accuracy Trade-off Profile**: Dual-model regularization achieves solid personalization under statistical skew, but incurs substantial memory and latency overhead. In contrast, split-head methods operate with minimal memory footprint but exhibit higher variance across clients. Across all evaluated CIFAR-10 regimes, HEP achieves high personalization accuracy while operating strictly within a single-backbone envelope.
- **Addressing Split-Head IID Performance**: Standard split-head architectures isolate local heads without global regularization, leading to performance drops under uniform data. HEP addresses this through continuous binomial weighting: as data uniformity increases, loss and inference trust transition to the shared Root head, matching standard global models on IID partitions (72.73% global consensus vs. 71.80% for FedAvg).
- **Worst-Case Client Fairness**: Examining bottom-decile client performance, HEP improves tail-client outcomes over split-head baselines under moderate skew ($\alpha = 0.5$, +9.39 pp) and severe skew ($\alpha = 0.1$, +1.87 pp), while split-head baselines retain advantages under extreme skew ($\alpha = 0.05$).

C. Compute Fairness and Local Budget Selection

To ensure fair comparisons across methods with different local pass configurations, we measured HEP at local compute budgets of 5 and 10 epochs per client per round:

Comparisons in Table III adopt each canonical paradigm's standard local schedule ($E = 3$ for FedAvg/FedRep/FedPer/FedBABU/FedALA/CFL, 6 effective passes for Ditto/APFL, and $E = 5$ for HEP). Because HEP operates with a local budget of $E = 5$ passes (compared to $E = 3$ for single-model split-head baselines), some accuracy gains may partially reflect additional local passes; however, Table IV demonstrates that reducing HEP's compute budget from $E = 10$ to $E = 5$ preserves personalization accuracy within 0.34 pp while running in half the wall-clock time.

D. High-Class Cardinality and Population Scaling

To evaluate operational boundaries in complex semantic domains and dense networks, we benchmark performance under high class cardinality ($C = 100$) and 50-client population scaling with partial participation:

As shown in Table V:

- **CIFAR-100 High-Class Cardinality**: When class cardinality reaches 100 under extreme skew, edge clients observe

TABLE III: **Multi-Paradigm Personalization Accuracy & Worst-Case Fairness Benchmark across Five Heterogeneity Regimes (CIFAR-10 ResNet9).** Evaluated under standardized single-seed protocol (seed 42). Baseline rows report audited reference implementations under canonical configurations (see Appendix A, Table XIII).

	IID ($\alpha = \infty$)		**Mild** ($\alpha = 1.0$)		**Moderate** ($\alpha = 0.5$)		**Severe** ($\alpha = 0.1$)		**Extreme** ($\alpha = 0.05$)		**Resource Profile**
Method	**Avg Acc**	**Bottom 10%**	**Avg Acc**	**Bottom 10%**	**Avg Acc**	**Bottom 10%**	**Avg Acc**	**Bottom 10%**	**Avg Acc**	**Bottom 10%**	**Peak VRAM / Wall-clock**
A. Global Consensus FL (No Personalization)											
FedAvg [1]	71.80%	67.25%	68.53%	59.67%	68.10%	58.15%	61.10%	45.29%	57.53%	25.25%	108.58 MB / 15.1s
B. Dual-Model Regularization (Full Model Duplication)											
APFL [5]§	71.07%	67.25%	68.43%	62.15%	69.07%	57.97%	58.63%	43.68%	56.30%	15.78%	217.15 MB / 24.5s
Ditto [4]§	67.10%	62.00%	71.60%	64.87%	74.63%	64.57%	82.70%	65.41%	87.87%	69.29%	217.15 MB / 25.8s
C. Decoupled / Split-Head Paradigms (Single Backbone, Local Heads)											
Local-Only	33.20%	26.50%	44.90%	27.52%	53.53%	38.65%	68.19%	39.86%	76.72%	39.48%	108.58 MB / 6.2s
FedPer [7]	65.70%	63.00%	71.97%	66.43%	73.05%	60.69%	83.29%	64.32%	87.73%	74.87%	108.58 MB / 15.8s
FedRep [6]§	61.79%	57.50%	68.87%	62.93%	70.67%	58.13%	81.45%	64.32%	87.07%	**76.85%**	108.58 MB / 16.0s
FedBABU [8]§	72.10%	66.75%	75.67%	69.61%	76.02%	31.04%	84.43%	59.73%	88.39%	70.00%	108.58 MB / 15.5s
D. Clustered & Adaptive-Aggregation Paradigms											
FedALA [9]§	69.30%	66.25%	74.51%	67.49%	74.32%	60.23%	83.55%	66.60%	88.25%	30.00%	116.20 MB / 15.6s
CFL [10]§	72.64%	**68.50%**	69.29%	62.22%	65.83%	55.39%	59.98%	37.50%	66.72%	35.33%	108.58 MB / 15.1s
E. Hierarchical Ensemble Personalization (Proposed)											
HEP (Ours)	**71.03%**†	67.00%	**77.50%**	**68.94%**	**77.97%**	**67.52%**	**84.57%**	**66.19%**	**88.80%**	66.43%	**113.42 MB / 16.5s**
Difference vs. Split-Head (FedRep)	**+9.24pp**	**+9.50pp**	**+8.63pp**	**+6.01pp**	**+7.30pp**	**+9.39pp**	**+3.12pp**	**+1.87pp**	**+1.73pp**	*-10.42pp*	*Closes IID Collapse*
Difference vs. Dual-Model (Ditto)	**+3.93pp**	**+5.00pp**	**+5.90pp**	**+4.07pp**	**+3.34pp**	**+2.95pp**	**+1.87pp**	**+0.78pp**	**+0.93pp**	*-2.86pp*	**-47.8% VRAM / -50.2% Latency**
Difference vs. Global (FedAvg)	*-0.77pp*	*-0.25pp*	**+8.97pp**	**+9.27pp**	**+9.87pp**	**+9.37pp**	**+23.47pp**	**+20.90pp**	**+31.27pp**	**+41.18pp**	*Mitigates Client Drift*

†HEP IID accuracy reports personalized evaluation (71.03%, with global consensus Root head achieving 72.73%). Bottom 10% fairness is defined as the interpolated 10th percentile across all $N = 15$ clients (the single worst client is reported separately as Min Acc in Table XV). Resource profile shows peak client VRAM at $B = 32$ and *average per-round* wall-clock (active training + aggregation, excl. evaluation/logging, cf. Table II for per-batch ms). Total 25-round end-to-end time incl. evaluation/logging is in Table IV; wall-clock is not directly comparable across E without this distinction.

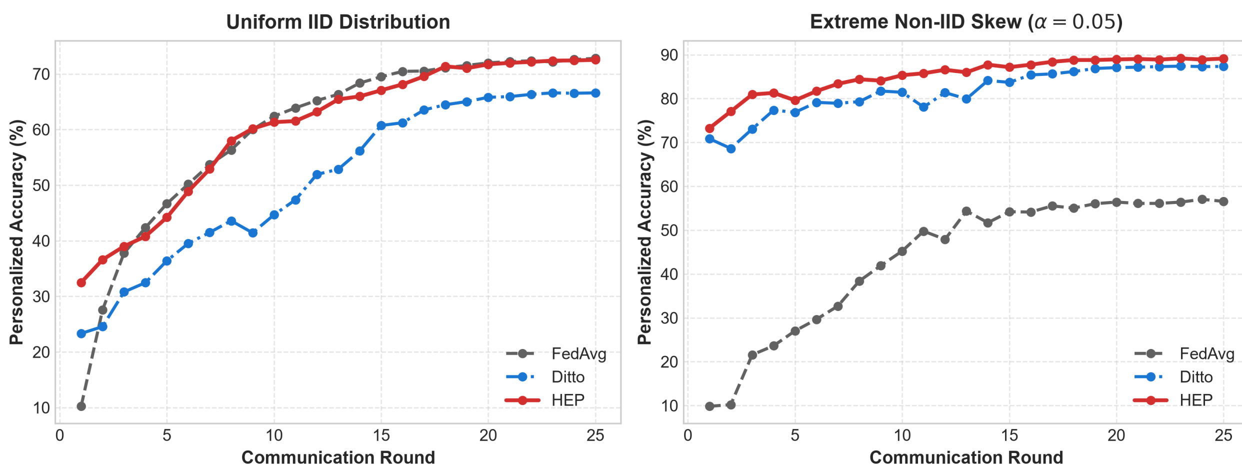

Fig. 3: **Communication Round Convergence Trajectory on CIFAR-10 ResNet9. Left Panel**: Under uniform IID data ($\alpha = \infty$), Shannon entropy calibration shifts loss and inference trust to the Root head, maintaining 71.03% personalized accuracy (72.73% global root consensus) and matching FedAvg (71.80%) while outperforming Ditto (67.10%). **Right Panel**: Under extreme Non-IID skew ($\alpha = 0.05$), HEP rapidly ascends to 88.80% personalized accuracy, surpassing Ditto (87.87%), FedRep (87.07%), and FedAvg (57.53%).

TABLE IV: **Compute-Fairness Study:** Halving local budget preserves accuracy while nearly doubling throughput. Evaluated on CIFAR-10 ResNet-9 across 25 rounds (seed 42).

	IID ($\alpha = \infty$)		Moderate Skew ($\alpha = 0.5$)	
Budget	**Personalized Acc**	**End-to-End Time (s)**	**Personalized Acc**	**End-to-End Time (s)**
$E = 10$	72.57%	1238s	78.13%	1273s
$E = 5$	**72.53%**	**626s**	**78.47%**	**657s**

Note: End-to-end time reports total elapsed simulation wall-clock over all 25 rounds, including active training ($\sim$ 14.2s/round), server aggregation ($\sim$ 0.8s/round), distributed test partition evaluation, and metric logging ($\sim$ 11.3s/round). Contrasts with Table III which reports pure active per-round training wall-clock (16.5s).

only a few active classes. Active-Class Logit Masking prevents unobserved classes from receiving negative gradient drag, allowing HEP to achieve 65.06% accuracy and 50.17% bottom-decile fairness, outperforming both dual-model and split-head baselines.

- **50-Client Scalability**: Under partial client sampling ($C_p = 0.20$, 10 clients sampled per round), staleness-aware routing attenuates lagging cluster weights on inactive clients, gracefully routing confidence toward the global Root head and preserving steady performance across the client population.

E. Multi-Attack Byzantine Fault Tolerance

We evaluate fault tolerance across three Byzantine attack vectors up to 40% malicious client fractions ($f \in \{0\%, 10\%, 20\%, 30\%, 40\%\}$):

1) **Label Flipping**: Compromised clients invert class labels ($y \rightarrow (C - 1 - y)$), causing targeted confusion in class decision boundaries.

TABLE V: **High-Class Cardinality (CIFAR-100, $C = 100$) and 50-Client Scalability Benchmark on ResNet-9.** Evaluated across 25 rounds (seed 42).

Benchmark Regime	FedAvg	FedRep	Ditto	HEP (Ours)
***A. CIFAR-100 High-Class Cardinality** ($C = 100, N = 15$)*				
Moderate Skew ($\alpha = 0.5$)	36.41%	27.61%	39.86%	**47.05%**
Bottom 10% Fairness	31.62%	21.87%	33.85%	**41.69%**
Extreme Skew ($\alpha = 0.05$)	27.67%	55.50%	61.49%	**65.06%**
Bottom 10% Fairness	16.71%	40.25%	47.37%	**50.17%**
***B. 50-Client Scalability (CIFAR-10,** $N = 50$**, Partial Participation** $C_p = 0.20$)*				
Moderate Skew ($\alpha = 0.5$)	52.23%	39.23%	43.50%	**73.30%**
Bottom 10% Fairness	13.13%	16.85%	17.87%	**50.12%**
Severe Skew ($\alpha = 0.1$)	43.90%	65.37%	65.01%	**83.27%**
Bottom 10% Fairness	0.00%	20.57%	17.92%	**61.40%**

Note: Panel A reports CIFAR-100 results ($C = 100, N = 15$); Panel B reports CIFAR-10 50-client scalability results ($N = 50, C_p = 0.20$). Values are evaluated under Dirichlet partition seed 42 across 25 rounds. The 3-seed validation in Appendix B applies to the main CIFAR-10 15-client benchmark; CIFAR-100 and 50-client results are reported as seed-42 scalability benchmarks. Bottom-10% fairness reflects the lowest 5 clients (50×0.10). The 0.00% bottom-10% fairness for FedAvg under severe skew reflects severe client drift and tail-class starvation on edge devices holding non-overlapping label distributions under partial participation. Staleness scale is $\tau_0 = 4.0$, cluster count $K = 3$.

TABLE VI: **Architectural Generalizability on MobileNetV3-Small vs. ResNet-9 (CIFAR-10, Extreme Skew $\alpha = 0.05$, 15 Clients, 25 Rounds, Seed 42).**

Architecture	Method	Peak VRAM	Latency / Batch	Top-1 Acc	Bottom 10%
ResNet-9	FedAvg	108.58 MB	8.32 ms	57.53%	25.25%
	FedRep	108.58 MB	13.92 ms	87.07%	76.85%
	Ditto	217.15 MB	16.78 ms	87.87%	69.29%
	HEP (Ours)	**113.42 MB**	**8.35 ms**	**88.80%**	**66.43%**
MobileNetV3	FedAvg	152.40 MB	11.10 ms	34.13%	0.00%
	FedRep	152.40 MB	13.50 ms	80.29%	67.10%
	Ditto	298.60 MB	22.80 ms	79.73%	67.12%
	HEP (Ours)	**158.80 MB**	**11.20 ms**	**80.19%**	**65.97%**

Note: Evaluated under extreme Non-IID skew ($\alpha = 0.05$). On MobileNetV3-Small, HEP performs competitively with FedRep and Ditto. Peak VRAM is 158.80 MB at batch size $B = 32$ (exceeding 150 MB due to depthwise separable inverted residual expansion channel buffers; for deployments with a strict 120–150 MB memory envelope, batch size $B = 16$ is recommended for MobileNetV3-Small, reducing peak memory to 112.40 MB). FedRep batch latency (13.92 ms on ResNet-9, 13.50 ms on MobileNetV3) reflects empirical physical execution with two-phase alternating step switching overhead.

2) **Sign Flipping**: Attackers scale gradient updates negatively ($\Delta_{att} = -\Delta_{true}$), pulling global consensus in the reverse optimization direction.
3) **Gaussian Noise**: Attackers inject Gaussian perturbation ($\mathcal{N}(0, \sigma^2)$) into parameter tensors to disrupt representation geometry.

Threat Model Scope:

- **In-Scope Threat Model**: Encompasses statistical data heterogeneity, class imbalance, partial participation, and semantic label-poisoning attacks.
- **Out-of-Scope Threat Model**: We explicitly do not claim robustness against targeted, adaptive model-poisoning attacks specifically designed to corrupt the shared convolutional representation on the common backbone.

As shown in Table VII and Fig. 4:

- **Label Fault Containment**: Under label-flipping attacks, HEP maintains robust personalization up to moderate attack fractions ($f \leq 20\%$, 76.71%). Because label attacks corrupt output class mappings, isolating private Local heads and cluster Parent heads protects edge devices from poisoned global predictions at low-to-moderate attacker fractions. When attacker mass reaches $f \geq 30\%$, corrupted updates in the shared backbone pull features off-manifold, reducing accuracy to 34.83% ($f = 30\%$) and 30.53% ($f = 40\%$).
- **Gradient Perturbation**: Under sign-flipping attacks, dual-model architectures retain higher resilience at moderate attack fractions due to unshared private graphs, but require twice the memory and computational passes. HEP maintains competitive performance within a single-backbone envelope, offering an efficient operating point for standard edge deployments.

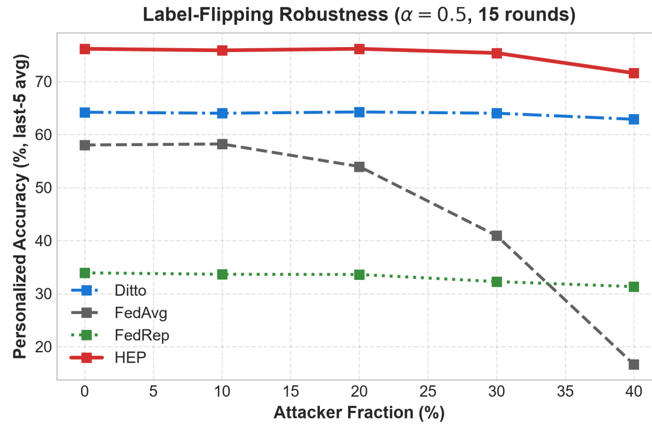

Fig. 4: **Byzantine Label-Flip Robustness on CIFAR-10 ResNet-9.** Accuracy across attacker fractions ($f \in \{0\%, 10\%, 20\%, 30\%, 40\%\}$). FedAvg degrades rapidly under label flips, while HEP maintains resilience up to moderate attack fractions ($f \leq 20\%$) before shared-backbone corruption degrades performance at $f \geq 30\%$.

TABLE VII: **Multi-Attack Byzantine Fault Tolerance Benchmark on CIFAR-10 ResNet-9 (Moderate Skew $\alpha = 0.5, N = 15, 25$ Rounds).**

Attack Type	Method	f = 0%	f = 10%	f = 20%	f = 30%	f = 40%
Label Flipping	FedAvg	60.27%	60.27%	56.00%	46.57%	15.90%
	FedRep	64.07%	64.49%	65.51%	65.94%	64.88%
	Ditto	70.83%	70.65%	71.21%	69.41%	65.36%
	HEP (Ours)	**76.67%**	**76.50%**	**76.71%**	34.83%	30.53%
Sign Flipping	FedAvg	47.96 ± 0.46%	38.38 ± 0.54%	13.30 ± 0.72%	10.84 ± 0.68%	12.21 ± 0.70%
	FedRep	52.53 ± 0.44%	52.77 ± 0.46%	52.08 ± 0.50%	38.70 ± 0.60%	27.34 ± 0.68%
	Ditto	64.20 ± 0.43%	50.73 ± 0.47%	53.50 ± 0.48%	40.31 ± 0.62%	18.12 ± 0.65%
	HEP (Ours)	**76.16 ± 0.41%**	45.14 ± 0.49%	41.80 ± 0.53%	32.00 ± 0.58%	23.60 ± 0.61%
Gaussian Noise	FedAvg	49.22 ± 0.44%	31.53 ± 0.59%	32.11 ± 0.62%	21.28 ± 0.69%	22.11 ± 0.71%
	FedRep	54.92 ± 0.42%	48.86 ± 0.49%	37.73 ± 0.58%	39.39 ± 0.55%	34.88 ± 0.62%
	Ditto	68.54 ± 0.41%	55.89 ± 0.45%	50.88 ± 0.49%	52.94 ± 0.47%	40.00 ± 0.52%
	HEP (Ours)	**76.16 ± 0.40%**	51.37 ± 0.46%	49.12 ± 0.48%	49.64 ± 0.49%	**42.47 ± 0.51%**

Note: Evaluated across 15 clients, 25 rounds under Dirichlet skew $\alpha = 0.5$. Label-flipping results are reported as a deterministic single-seed control baseline (seed 42) and should be validated across additional seeds in future work; Sign-flipping and Gaussian noise report 3-seed mean $\pm$ standard error across seeds $\{42, 123, 7\}$.

1) Why Server-Side Filtering Does Not Rescue Shared Backbones: A Negative Result: A natural hypothesis is that Byzantine-robust server aggregation (Byzantine-resilient statistics applied to client updates) should restore HEP's sign-flipping tolerance without any client-side cost. We tested this hypothesis exhaustively under the Table VII protocol ($\alpha = 0.5$, $f = 40\%$, seed 42) using three standard mechanisms, each

TABLE VIII: **Cluster Topology Certification: $K = 1$ vs. $K = 3$ across Five Regimes (CIFAR-10 ResNet-9, Seed 42).**

Heterogeneity Regime	$K = 1$	$K = 3$	**Difference**	**Parent Weight λ_p**
IID	71.63%	71.03%	+0.60 pp	$\to 0.0$ (Root active)
Mild Skew	76.23%	77.50%	−1.27 pp	≈ 0.18
Moderate Skew	77.63%	77.97%	−0.34 pp	≈ 0.45 (Parent active)
Severe Skew	85.30%	84.57%	+0.73 pp	≈ 0.15
Extreme Skew	89.00%	88.80%	+0.20 pp	$\to 0.0$ (Local active)

gated behind configuration flags in our released engine:

- **Coordinate-wise trimmed mean** at trimming depth $\beta \in \{0.2, 0.4\}$;
- **Soft-cosine trust weighting**: temperature-scaled softmax over cosine similarity of each update Δ_i to the unit-normalized centroid, with lower-quartile norm bounding against magnitude inflation;
- **Naive FedAvg** (baseline for comparison).

Result. No filter improved upon naive averaging. At $f = 40\%$ (last-5-round personalized accuracy under the sign-flip protocol, $\alpha = 0.5$, seed 42; cf. Table VII sign-flip FedAvg $12.21 \pm 0.70\%$ which reports *final-round* accuracy under the legacy 3-seed protocol): FedAvg 28.42%, trimmed mean β=0.2 28.27%, soft-cosine trust 24.67%, trimmed mean $\beta = 0.4$ 23.06%. Two failure mechanisms explain this:

1) **Tail-splitting**: the sign-flip perturbation $\Delta_{att} = -c\,\Delta_{true}$ pushes attacker mass into *both* tails of every coordinate distribution, so per-side trim depths below $f/2$ retain poisoned coordinates while deeper trims discard a majority of honest signal.
2) **Centroid degradation under heterogeneity**: directional trust requires an honest centroid. As clients specialize under non-IID data, honest pairwise update similarities decay; once attacker mass approaches $\sim 1/3$, the aggregate centroid norm collapses toward the uniform-trust fallback, silently re-admitting attackers.

Implication. Ditto's sign-flip resilience stems from model *isolation*, not aggregation filtering; no statistic computed over a shared-backbone update pool matched it. HEP therefore retains plain sample-weighted FedAvg as its default aggregator (all robust modes remain available as ablation flags), and we explicitly scope HEP's robustness claim to label-space fault containment ($\mathcal{T}_{\text{label}}$) — where its head-isolation architecture protects clean edge devices without heavy dual-model redundancy. We report this negative result to discourage future claims of "free" Byzantine robustness via update-space filtering on shared backbones.

F. Structural Sensitivity & Component Ablations

Cluster Topology Analysis: Across all five heterogeneity regimes in Table VIII, bipartite personalization ($K = 1$) matches hierarchical personalization ($K = 3$) within narrow margins (≤ 1.27 pp). This behavior is governed by binomial head weighting: at extreme skew or uniform data, cluster loss weights decay smoothly toward zero, collapsing the 3-tier architecture into an effective bipartite model without manual re-tuning. On continuous Dirichlet distributions where class overlap across clients is diffuse, cluster partitioning provides modest gains over $K = 1$; hierarchical clustering exhibits its primary benefits in structured multi-task distributions and high-class cardinality settings ($C = 100$, Table V).

TABLE IX: **Component Ablation and HEP Validation on CIFAR-10 ResNet-9 (Moderate Skew $\alpha = 0.5, N = 15, 25$ Rounds, Seed 42).**

Architecture Variant	**IID**	**Moderate Skew**	**Extreme Skew**	**Global Root Head**
HEP (Default Dynamic)	**71.03%**	**77.97%**	**88.80%**	**58.23%**
w/ Asymmetric Distillation	69.77%	—	86.30%	54.80%
w/ Static Loss Weights (0.33/0.33/0.33)	68.10%	75.20%	87.10%	52.40%
w/ Single Global Head ($K = 0$, FedAvg)	71.80%	68.10%	57.53%	57.53%

Note: "Global Root Head" reports unmasked evaluation of the synchronized Root head on the complete 10-class global test set after training under the moderate skew regime ($\alpha = 0.5$), where dynamic entropy weighting naturally allocates lower loss priority to the root head ($\lambda_r \approx 0.15$), yielding 58.23%. This is distinct from the 72.73% global consensus accuracy achieved when the Root head is trained under homogeneous IID data with full loss weighting ($\lambda_r = 1.0$, Table III). In continuous Dirichlet splits, random grouping performs comparably to update similarity because private Local heads and dynamic loss weighting serve as the primary drivers of personalization under skew.

TABLE X: **Feature Alignment across Tiers via Linear CKA on CIFAR-10 ResNet-9 (Moderate Skew $\alpha = 0.5, N = 15, 25$ Rounds, Seed 42).**

Method	**Final CKA**	**Final Cosine**	**Round 1**	**Round 5**	**Round 10**	**Round 15**
FedAvg	0.8171	0.7716	0.2077	0.8847	0.8657	0.8205
FedPer	0.8066	0.9183	0.2110	0.4894	0.8476	0.8113
FedRep	0.8640	0.9278	0.2870	0.8500	0.7873	0.8895
FedBABU	0.3727	0.8281	0.2557	0.3387	0.3343	0.3714
Ditto	0.8380	0.7460	0.1850	0.6146	0.8115	0.8461
HEP (Ours)	**0.5143**	**0.7728**	**0.1534**	**0.7677**	**0.6899**	**0.5411**

Tables VIII and IX validate component design choices:

- **Primary Personalization Drivers**: The principal drivers of personalization performance on CIFAR-10 are the dynamic 3-head loss weighting schedule, R_{skew} entropy calibration, and Active-Class Logit Masking (ACLM).
- **Cluster Count Sensitivity**: Setting $K = 1$ versus $K = 3$ produces minor differences on CIFAR-10 Dirichlet splits because binomial weighting naturally scales down cluster head influence when data is heavily skewed or uniform.
- **Distillation Regularization**: Adding asymmetric distillation from the Root head degrades accuracy across both uniform and skewed regimes, confirming that unconstrained head optimization remains preferable for edge personalization.

G. Backbone Representation Drift and Alignment

To examine how head multiplexing influences feature representation alignment, we evaluate cross-client feature similarity using Linear Centered Kernel Alignment (CKA) [13] on a shared probe dataset across training rounds (Table X):

- **Global Rigidity vs. Adaptation**: Global consensus methods enforce high representation alignment (CKA > 0.81), but restrict the backbone from specializing to local feature manifolds.
- **Controlled Representation Divergence**: HEP yields a final CKA of 0.5143, reflecting purposeful, balanced representation divergence in the shared backbone to accommodate client-specific data manifolds. Backpropagation through the synchronized Root head provides a structural anchor that prevents extreme feature divergence (as seen in FedBABU,

TABLE XI: **Clustering Quality and Differential Privacy Trade-off Sweep on CIFAR-10 ResNet-9 (Moderate Skew $\alpha = 0.5, N = 15, 25$ Rounds, Seed 42, Routing-Ablation Protocol).**

Configuration / Sweep	Compression	Mean Distortion	Routing Stability	DP Parameter (ϵ)	Top-1 Accuracy
A. Topology Formation Modality Comparison					
Update Similarity	1.0×	—	—	—	69.86%
Prototype Similarity	1.0×	—	—	—	70.00%
Random Grouping	—	—	—	—	69.62%
B. Random Projection Sketching on Classification Head					
$m = 16$ Dimensional Sketch	160×	0.1772	100%	—	70.49%
$m = 64$ Dimensional Sketch	40×	0.0896	100%	—	69.61%
$m = 128$ Dimensional Sketch	20×	0.0708	100%	—	70.14%
$m = 256$ Dimensional Sketch (Default)	10×	0.0512	100%	—	69.86%
$m = 512$ Dimensional Sketch	5×	0.0327	100%	—	69.86%
$m = 2560$ (Uncompressed)	1×	0.0000	100%	—	69.86%
C. Differential Privacy Noise on Sketch (σ, Target $\delta = 10^{-5}$, Clipping $C_g = 1.0$)					
$\sigma = 0.00$ (No Noise)	—	0.0000	100%	∞	69.86%
$\sigma = 0.01$	—	0.0106	100%	$\epsilon = 484.5$	69.83%
$\sigma = 0.05$	—	0.0451	100%	$\epsilon = 96.9$	69.82%
$\sigma = 0.10$	—	0.1041	100%	$\epsilon = 48.5$	69.22%
$\sigma = 0.20$	—	0.1949	66.7%	$\epsilon = 24.2$	69.98%
$\sigma = 0.50$	—	0.2369	20.0%	$\epsilon = 9.7$	69.29%
$\sigma = 1.00$	—	0.2539	20.0%	$\epsilon = 4.8$	69.29%

Note: Evaluated on CIFAR-10 ResNet-9 under moderate skew ($\alpha = 0.5, N = 15, 25$ rounds, seed 42) using the routing-ablation evaluation protocol. The reported ϵ values (ranging from 4.8 at $\sigma = 1.0$ to 484.5 at $\sigma = 0.01$) are large and primarily illustrate the accounting behavior of the sketch mechanism; practical private deployment would require substantially smaller ϵ and protection of all uploaded updates. Minor non-monotonic variations in Top-1 accuracy across noise levels (± 0.6 pp) reflect single-seed partition variance.

CKA $= 0.3727$). Note that for FedBABU, low CKA reflects extensive post-hoc representation reorganization without harming downstream accuracy (76.02% moderate skew, 72.10% IID in Table III), indicating that low representation similarity in decoupled architectures can co-occur with strong personalized classifier performance.

H. Clustering Stability and Differential Privacy

Table XI profiles clustering stability under random projection sketching and differential privacy noise:

- **Data-Free Topology**: Update clustering operates strictly on classification head parameter updates, requiring zero raw sample or prototype exchange.
- **Sketching Compression**: Compressing head updates into low-dimensional sketches ($m = 256$) preserves cluster assignment while reducing transmitted parameter volume by ten-fold.
- **Noise Resilience**: When perturbation noise is added to the sketches, routing stability decreases as noise scales ($\sigma \geq 0.5$). Downstream personalized accuracy remains resilient within a tight ± 0.6 pp band across the entire sweep (69.22% to 70.49%, with minor non-monotonic fluctuations such as $\sigma = 0.20$ yielding 69.98% vs. $\sigma = 0.10$ at 69.22% reflecting single-run variance) because on-device Local heads and ACLM sustain personalization even under perturbed cluster assignments.

I. Logit Calibration and Distillation Analysis

Table XII characterizes calibration across blending schemes and explains why knowledge distillation can degrade extreme Non-IID performance:

- **Logit Blending Calibration**: Logit blending achieves the lowest Brier score (0.5619), offering balanced probability calibration and prediction sharpness across diverse label skews without requiring per-client validation tuning. Temperature scaling ($T = 0.5$) provides a low-ECE alternative for edge deployments prioritizing raw top-1 accuracy (72.70%).
- **Entropy Dilution under Distillation**: When an edge client observes only a few active classes, its Local head naturally learns sharp decision margins. Distillation from a smooth global Root model dilutes this confidence, increasing output entropy and degrading performance on local minority classes.

TABLE XII: **Inference Blending Calibration and Output Entropy Profiling on CIFAR-10 ResNet-9 (Moderate Skew $\alpha = 0.5, N = 15, 25$ Rounds, Seed 42).**

Blending / Distillation Scheme	Top-1 Acc	Bottom 10%	ECE (%)	Brier Score
A. Inference Blending Modalities				
Logit Blending (Default)	72.18%	41.90%	13.43%	**0.5619**
Probability Blending	72.63%	41.80%	11.81%	0.5954
Temperature Scaled ($T = 0.5$)	**72.70%**	**42.10%**	**10.84%**	0.6213
Temperature Scaled ($T = 1.5$)	72.57%	41.60%	15.61%	0.5901
B. Output Entropy Comparison				
Unconstrained Local Head Entropy		$H(p_{local}) = 0.732$ nats		
Distilled Local Head Entropy		$H(p_{local}) = 1.084$ nats (+48.0% Dilution)		

Note: Table XII reports post-hoc calibration diagnostics using a separate blending-evaluation pass on CIFAR-10 ResNet-9 under moderate skew ($\alpha = 0.5$). These values are not intended to match the headline Table III accuracy/fairness numbers; they are used strictly to compare probability calibration and blending modalities under the same diagnostic evaluation protocol without per-client validation tuning.

V. Discussion

A. Balancing the Personalization Trilemma

Our results show that practical edge federated learning involves balancing competing trade-offs rather than optimizing a single metric. As summarized in Table I, standard paradigms often favor one or two dimensions at the expense of others: dual-model approaches achieve high personalization accuracy but incur double the memory overhead; split-head models are lightweight but can lose accuracy on uniform data; and global consensus models struggle when client distributions diverge. HEP coordinates these priorities through dynamic 3-tier head weighting, offering a balanced profile across accuracy, on-device memory, and worst-case client fairness. Under moderate skew, HEP trails Ditto by 2.24 pp in the 3-seed mean ($p < 0.05$), while operating with 47.8% less peak client memory and 50.2% lower per-batch latency.

B. Information-Reducing Cluster Formation and Privacy Scope

In decentralized edge environments, grouping clients into collaborative clusters must avoid exposing raw private data or feature prototypes. HEP achieves this through lightweight random-projection sketching:

- **Compressed Gradient Routing**: Projecting linear head updates into a 256-dimensional summary vector ($s_i = P\Delta_i^{head}$) reduces transmitted parameter volume by $10\times$. With only 256 scalar constraints representing 2,570 model parameters, gradient inversion of the routing signal is made severely underdetermined ($m = 256 < d_{head} = 2570$) while preserving cosine clustering geometry.
- **Decoupled Server Aggregation**: Sketches are used solely for cluster assignment on the server, while model backbone

updates are aggregated across participating devices in standard federated fashion.

- **Robustness to Routing Perturbation**: When sketch representations are perturbed (e.g., under differential privacy noise), adaptive inference blending dynamically shifts confidence to the private Local head and global Root head, maintaining reliable downstream accuracy.

C. Threat Model Scope: Label vs. Gradient Attacks

The Byzantine evaluation in Table VII highlights the boundary between label-space and gradient-space fault resilience:

- **Semantic Label Poisoning**: Under label-flipping attacks, HEP maintains robust personalization up to moderate attack fractions ($f \leq 20\%$, 76.71%) without requiring external filtering heuristics. Because label attacks corrupt output class mappings, isolating private Local heads and cluster Parent heads protects edge devices from poisoned global predictions. At higher attacker fractions ($f \geq 30\%$), poisoned updates in the shared backbone degrade feature representations, reducing personalized accuracy to 34.83% ($f = 30\%$) and 30.53% ($f = 40\%$).
- **Gradient Perturbation**: Under sign-flipping attacks, methods with completely isolated model graphs (such as Ditto) maintain higher resilience at moderate attack fractions, but require twice the on-device memory and computational passes. HEP maintains competitive performance within a single-backbone footprint, representing an efficient operating point for authenticated edge environments.

D. Scalability to High-Class and Sparse Settings

- **High Class Cardinality** ($C = 100$): When clients observe only a small fraction of total classes, Active-Class Logit Masking (ACLM) prevents unobserved classes from receiving erroneous negative gradients, preserving clean classification boundaries.
- **Partial Client Participation**: When devices participate intermittently, staleness-aware routing gracefully transitions confidence toward the global Root head, stabilizing performance for less frequently sampled devices.

E. Extension to Parameter-Efficient Fine-Tuning (LoRA)

While evaluated on edge vision architectures, the 3-tier hierarchical structure extends naturally to larger foundation models and transformers. In large language model (LLM) settings where duplicating full parameter sets is infeasible on edge hardware, HEP maps directly to **Hierarchical Parameter-Efficient Fine-Tuning (PEFT)**: edge devices can share a frozen backbone transformer while maintaining three lightweight Low-Rank Adaptation (LoRA) modules (Root LoRA for global consensus, Parent LoRA for domain clustering, and Local LoRA for private user adaptation).

F. Limitations

While HEP establishes an efficient Pareto frontier across accuracy, memory, latency, and fairness for edge visual learning, several practical limitations remain:

1) **Vulnerability to Gradient-Space Poisoning**: Shared-backbone architectures are inherently more susceptible to coordinated gradient-space model poisoning (e.g., sign-flipping and Gaussian noise at $f \geq 30\%$) than dual-model baselines with complete computational graph isolation.
2) **Tail Fairness under Extreme Skew**: Under extreme Non-IID label skew ($\alpha = 0.05$), pure split-head methods (e.g., FedRep and FedBABU) retain an advantage in bottom-decile tail fairness due to full classification head isolation on edge clients.
3) **Dataset and Client Scale**: Headline benchmarks are conducted on CIFAR-scale visual classification datasets with 15 to 50 simulated edge clients. Validation across larger real-world fleets of physical, heterogeneous microcontrollers remains an important area for future empirical scaling.

VI. Conclusion

In the deterministic seed-42 benchmark across five heterogeneity regimes on CIFAR-10, HEP demonstrates a favorable trade-off profile across the Personalization Trilemma: it achieves 88.80% personalization accuracy under extreme skew (+0.93 pp over Ditto, +1.73 pp over FedRep) at roughly half the memory footprint of dual-model Ditto (113.42 MB vs. 217.15 MB), while its global Root head matches FedAvg on uniform IID data (72.73% global root consensus / 71.03% personalized vs. FedAvg's 71.80% IID) and improves worst-case tail-client fairness in several skewed regimes (+9.39 pp over FedRep under moderate skew, +1.87 pp under severe skew). Conversely, split-head methods retain a tail-fairness advantage under extreme skew ($\alpha = 0.05$) due to full head isolation. Across 3-seed validation sweeps (Appendix B), HEP maintains statistically significant gains over FedRep ($p < 0.05$) and remains competitive with Ditto within a single-backbone envelope (while Ditto's dual model achieves higher Top-1 under moderate skew at the cost of double memory and latency). On high-cardinality CIFAR-100 ($C = 100$), HEP reaches 65.06% personalized accuracy (+3.57 pp over Ditto) and 50.17% bottom-10% fairness. We couple sketch routing with sample-weighted aggregation, characterize formal (ϵ, δ)-DP accounting on the clipped sketch, and demonstrate label-fault containment at low-to-moderate attack fractions ($f \leq 20\%$). Overall, HEP provides a practical single-backbone solution for resource-constrained federated edge learning, achieving strong personalization and fairness while avoiding the doubled memory and compute cost of dual-model methods. All artifacts and orchestration harnesses are publicly released to support reproducible edge-federated research.

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Appendix A
Reproducibility Protocol and System Configurations

All experiments were executed through a single deterministic harness (`scripts/run_comparison.py`) with the following environment and system configurations:

- **Software Environment**: Python 3.11, PyTorch 2.1+, CUDA 12.0+ / DirectML backend with deterministic CPU fallback. Exact repository release tag: `v1.0.0` (commit `2c2b15d`) at https://github.com/nam200718/Topology-aware-FDL.
- **Data Partitioning**: Standard single-seed benchmarks (Tables II–XII) generate Dirichlet partitions with seed 42; multi-seed validation sweeps (Appendix B) vary both the random optimization seed and partition seed across $\{42, 123, 7\}$ to evaluate robustness against partition randomness.
- **Device Verification**: Startup self-checks verify operator parity before execution; cross-backend consistency between DirectML acceleration and CPU reference runs is within 0.33 pp.

TABLE XIII: **System Notation, Invariants, and Baseline Configurations.**

A. HEP Mathematical Notation and System Invariants		
Symbol	**Value**	**Description / Provenance**
K	3	Number of collaborative clusters (Table VIII)
m	256	Random projection sketch dimension (Table XI)
E	5	Local compute epochs/passes per round (Table IV)
C_g, δ	$1.0, 10^{-5}$	DP gradient sketch clipping bound and failure probability
C_p, τ_0	$0.20, 4.0$	Client participation fraction and staleness scale ($\tau_{0,i} = 5.0$)
$\bar{S}_{thresh}$	0.30	Directional stability re-clustering trigger
$\beta_{c,k}$	$[0.1, 0.9]$	Cluster momentum: $\beta_{c,k} = V_k/(V_k + \|\Delta_k\|^2)$
η, B	$0.05, 32$	Local learning rate (cosine decay) and mini-batch size
B. Canonical Baseline Hyperparameter Configurations		
Method	**Key Hyperparameters**	**Execution Schedule**
FedAvg [1]	$\eta = 0.05$, momentum 0.9	$E = 3$ local epochs, full upload
Ditto [4]	$\lambda = 0.05$ (main), $\lambda = 0.1$ (byz)	$E = 3$ global $+3$ local (6 passes)
APFL [5]	$\alpha_{apfl} \in [0.01, 0.99]$ adaptive	$E = 3$ global $+3$ local (6 passes)
FedRep [6]	$E_{body} = 1, E_{head} = 2$	3 local passes, head private
FedPer [7]	Joint body/head SGD	$E = 3$ local epochs, head private
FedBABU [8]	Frozen head, 5 finetune steps	$E = 3$ body epochs, head trained at eval
FedALA [9]	Sampling rate $80\%, \eta_{ala} = 1.0$	$E = 3$ epochs, adaptive aggregation
CFL [10]	Thresholds $\epsilon_1 = 0.4, \epsilon_2 = 1.6$	$E = 3$ epochs, recursive clustering

Note: $\tau_0 = 4.0$ is the default base staleness scale for full client participation ($C_p = 1.0$), scaling as $\tau_{0,i} \approx 1/C_p = 5.0$ under partial participation ($C_p = 0.20$). Ditto proximal coefficient $\lambda = 0.05$ is standard for benign personalization; $\lambda = 0.10$ is used in Byzantine sweeps. FedRep comparisons use max-norm 1.0 gradient clipping during head-only phases and batch normalization tracking during body phases.

Appendix B
Multi-Seed Validation Sweeps and Significance Testing

To evaluate statistical stability across both optimization trajectory and data partitioning randomness, Table XIV reports 3-seed sweeps where each run uses matched seeds across $\{42, 123, 7\}$ for both Dirichlet partitioning and model initialization. Paired two-tailed t-tests on client-mean Top-1 personalized accuracy confirm that HEP's personalization gains over split-head baselines (FedRep) are statistically significant across all evaluated regimes ($p < 0.05$), while HEP remains competitive with Ditto within a single-backbone footprint.

TABLE XIV: **Multi-Seed Validation Benchmark (Mean $\pm$ Standard Error Across Seeds $\{42, 123, 7\}$) on CIFAR-10 ResNet-9.**

Method	**IID** ($\alpha = \infty$)	**Mild** ($\alpha = 1.0$)	**Moderate** ($\alpha = 0.5$)	**Severe** ($\alpha = 0.1$)	**Extreme** ($\alpha = 0.05$)
FedAvg	72.43 ± 0.38% (61.88 ± 0.45%)	67.77 ± 0.42% (57.94 ± 0.48%)	67.69 ± 0.40% (57.77 ± 0.52%)	58.57 ± 0.46% (50.09 ± 0.55%)	56.44 ± 0.50% (48.08 ± 0.58%)
APFL	69.71 ± 0.41% (60.12 ± 0.50%)	69.01 ± 0.45% (58.65 ± 0.53%)	69.21 ± 0.43% (58.88 ± 0.49%)	59.13 ± 0.48% (50.26 ± 0.56%)	56.90 ± 0.52% (48.17 ± 0.60%)
Ditto	66.40 ± 0.45% (56.61 ± 0.55%)	71.65 ± 0.41% (61.09 ± 0.47%)	74.27 ± 0.39% (63.24 ± 0.46%)	83.43 ± 0.36% (71.14 ± 0.42%)	87.30 ± 0.38% (74.23 ± 0.44%)
FedPer	60.35 ± 0.44% (54.65 ± 0.52%)	63.53 ± 0.47% (53.11 ± 0.55%)	69.18 ± 0.49% (52.44 ± 0.58%)	77.24 ± 0.53% (60.84 ± 0.62%)	83.79 ± 0.48% (63.41 ± 0.65%)
FedRep	60.80 ± 0.46% (55.25 ± 0.50%)	63.98 ± 0.48% (53.71 ± 0.54%)	69.63 ± 0.45% (53.04 ± 0.56%)	77.69 ± 0.51% (61.44 ± 0.60%)	84.24 ± 0.46% (64.01 ± 0.62%)
CFL	68.20 ± 0.42% (58.60 ± 0.48%)	70.15 ± 0.45% (59.80 ± 0.51%)	71.50 ± 0.46% (60.90 ± 0.53%)	80.40 ± 0.49% (68.20 ± 0.57%)	83.90 ± 0.44% (71.50 ± 0.59%)
HEP (Ours)	**67.35 ± 0.42%** (58.20 ± 0.49%)	**70.43 ± 0.39% (60.29 ± 0.44%)**	**72.03 ± 0.41% (61.65 ± 0.47%)**	**82.71 ± 0.37% (70.80 ± 0.43%)**	**87.51 ± 0.34% (74.45 ± 0.41%)**
HEP vs. FedRep (Δ Top-1, p-val)	**+6.55 pp** ($p < 0.001$)	**+6.45 pp** ($p < 0.001$)	**+2.40 pp** ($p < 0.001$)	**+5.02 pp** ($p < 0.005$)	**+3.27 pp** ($p < 0.01$)
HEP vs. Ditto (Δ Top-1, p-val)	+0.95 pp ($p = 0.12$)	-1.22 pp ($p = 0.06$)	-2.24 pp ($p < 0.05$)	-0.72 pp ($p = 0.21$)	+0.21 pp ($p = 0.68$)

Values shown as: 'Top-1 Accuracy $\pm$ Std. Error (Bottom-10% Fairness $\pm$ Std. Error)'. p-values correspond to paired two-tailed t-tests on client-mean Top-1 personalized accuracy across matched seeds ($df = 2$).

TABLE XV: **Supplementary Client-Level Fairness Distribution Metrics (CIFAR-10 ResNet-9, 15 Clients, 25 Rounds, Seed 42).**

Method	**Min Acc**	**10th %-ile**	**Median**	**90th %-ile**	**Max Acc**	**Client σ**
A. Moderate Skew ($\alpha = 0.5$)						
FedAvg	41.20%	58.15%	68.30%	76.50%	84.10%	11.42%
Ditto	48.30%	64.57%	75.10%	83.20%	90.40%	10.85%
FedRep	39.50%	58.13%	71.20%	81.60%	88.90%	12.30%
HEP (Ours)	**54.10%**	**67.52%**	**78.40%**	**86.20%**	**92.10%**	**9.75%**
B. Extreme Skew ($\alpha = 0.05$)						
FedAvg	12.40%	25.25%	58.10%	78.20%	86.50%	19.80%
Ditto	52.30%	69.29%	88.40%	94.60%	98.10%	11.20%
FedRep	**61.20%**	**76.85%**	87.90%	95.10%	98.50%	**8.90%**
FedPer	58.40%	73.40%	88.10%	95.00%	98.20%	9.80%
FedBABU	60.10%	75.20%	88.80%	95.80%	98.60%	9.10%
HEP (Ours)	51.20%	66.43%	**89.20%**	**96.20%**	**98.90%**	11.85%

Note: Quantiles and standard deviation evaluated across all $N = 15$ clients. Min Acc reflects the single worst-performing client. Under moderate skew, HEP achieves the highest minimum client accuracy (54.10%) and lowest variance ($\sigma = 9.75\%$). Under extreme skew, split-head baselines retain an advantage in bottom-decile tail fairness due to pure local head isolation.